PREP 4, LESSONS 7–8

Using radiation and half-life

Ionising power, range and penetration; choosing a source; half-life from a graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard unit of activity. [1]

2 Describe how the mass number and atomic number change when an alpha particle is emitted. [1]

3 State what a beta particle is. [1]

Key ideas LESSONS 7–8

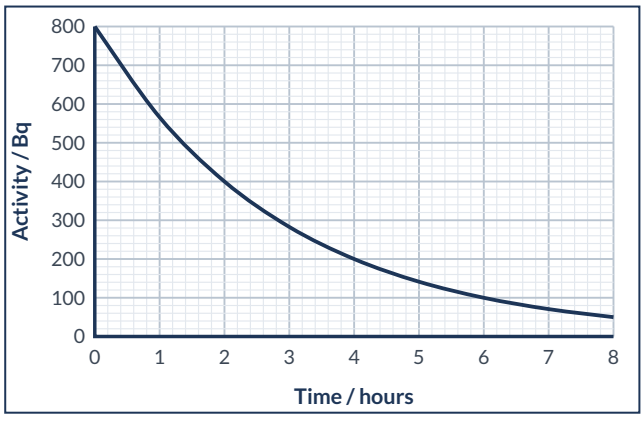
Show your working

4 The most ionising radiation is **alpha** | **beta** | **gamma** [1]

5 A factory uses a radioactive source and a detector to check the thickness of paper.

Draw a ring around the radiation it should use: **alpha** | **beta** | **gamma** Explain your choice. [2]

6 The graph shows how the activity of a source changes with time.



a Use the graph to find the half-life. Show on the graph how you did it. [2]

b Define half-life. [1]

c Radioactive decay is random. Explain what this means. [1]